

Course Syllabus

General Information

Faculty	Information Technology		
Department	Data Science and Artificial Intelligence		
Semester	2 st	Academic Year	2025/2026

Course Details

Course Title	Computer vision			Course No.	0434402		
Type of Learning	☒ On- campus ☐ Blended (..... Platform) ☐ Online (..... Platform)						
Credit Hours	3	Theoretical	2	Practical	1		
Course Level (according to JNQF)			7	JNQF Hours*		16.2	
Pre-requisite	Neural networks – 0433401 Neural networks Lab – 0433402			Co-requisite		N/A	

Course Instructor Information

Instructor Name	Maria Yousef		
Instructor E-mail	M.yousef@meu.edu.jo		
Course Time	(14:00-16:00)Sun, Tue		
Office Hours	9:00-10:00 , 13:00-14:00 (Sun, Tue, Thu) 13:00-14:00 (Mon, Wed)		
Office Number	N119	Office Phone	
Lab Instructor Name –If assigned-	N/A		

Sources and References

1. Textbook

Szeliski, R. (2020). Computer Vision: Algorithms and Applications (2nd ed.). Springer. ISBN: 978-3030343714

2. References

- Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press. ISBN: 978-0262035613.
- Gonzalez, R. C., & Woods, R. E. (2018). Digital Image Processing (4th ed.). Pearson. ISBN: 9780133356724.
- Forsyth, D. A., & Ponce, J. (2012). Computer Vision: A Modern Approach (2nd ed.). Pearson. ISBN: 9780136085928.

Teaching Methods

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|---------------------------|-----------------------------------|-------------------------------|
| 1. Lectures | 2. Discussion and problem-solving | 3. Interactive demonstrations |
| 4. Project-Based Learning | 5. Groups discussion | |

Course Description

This course provides a foundational understanding of computer vision, focusing on Python-based image processing and analysis techniques. Students will explore the representation and manipulation of visual data, covering image structures, raster and vector formats, and essential operations: filtering, and transformations. Advanced topics include feature detection, matching, and introductory object detection. By the end of the course, students will be equipped to implement core computer vision techniques and analyze visual data, laying the groundwork for more complex applications in the field.

Course Learning Outcomes (CLOs) and its correlation with Program Learning Outcomes (PLOs) and JNQF Descriptors

No.	CLOs	PLOs	JNQF Descriptors		
			Knowledge	Skill	Competency
1	Explain key concepts in computer vision, including image representation, manipulation, and feature extraction.	PLO1	✓		
2	Implement Python-based image processing techniques for transforming and analyzing visual data.	PLO1	✓		
3	Apply feature extraction and detection methods, including CNNs, for object recognition and classification.	PLO2		✓	
4	Build and evaluate CNN models and image classifiers to solve real-world image recognition tasks.	PLO2		✓	
5	Demonstrate ethical considerations in computer vision, addressing biases in image data and model predictions.	PLO4	✓		

Course Topics

Week	Topic	Lecture Type		CLOs	Assessment	
		Lecture Type	Lecture No.		Assessment Tool	Summative/ Formative
1	Course Syllabus discussion Introduction to Computer Vision: - Definition, evolution, and key applications. - Core Concepts: Image representation, formation, and vision pipeline.	Lecture (1)	Lecture On-campus	1	Discussion & Questions	Formative
		Lecture (2)	Lecture On-campus	1	Discussion & Questions	Formative
2	Understanding Digital Image: - image types (raster/vector), and key properties. - Image Processing: Resolution, color depth, and format conversion.	Lecture (1)	Lecture On-campus	1,2	Discussion & Questions	Formative
		Lecture (2)	Lecture On-campus	1,2	Discussion & Questions	Formative
3	- Python for Computer Vision & Image Processing Basics - Environment setup (OpenCV, NumPy, Matplotlib)	Lecture (1)	Lecture On-campus	1,2	Discussion & Questions	Formative
		Lecture (2)	Lecture On-campus	1,2	Discussion & Questions	Formative
4	- Image Manipulation: Loading, resizing. - cropping images; basic transformations (rotation, scaling) - Practical implementation of transformations and visualization using Python.	Lecture (1)	Lecture On-campus	1,3	Discussion & Questions	Formative
		Lecture (2)	Lecture On-campus	1,3	Discussion & Questions	Formative
5	Thresholding and Edge Detection - Edge Detection Techniques. - Fundamentals of edge detection and their role in identifying image boundaries. - Introduction to edge detection; Sobel and Prewitt methods.	Lecture (1)	Lecture On-campus	1,2,3	Quiz#1	Summative
		Lecture (2)	Lecture On-campus	1,2,3	Discussion & Questions	Formative

6	Feature Extraction Techniques Introduction to LBP and HOG; understanding texture and gradient-based features.	Lecture (1)	Lecture On-campus	2,4	Discussion & Questions	Formative
		Lecture (2)	Lecture On-campus	3	Discussion & Questions	Formative
7	Introduction to segmentation basic thresholding methods (binary, adaptive, Otsu's).	Lecture (1)	Lecture On-campus	4,5	Discussion & Questions	Formative
		Lecture (2)	Lecture On-campus	4,5	Discussion & Questions	Formative
8	Mid Exam			1,2,3	Mid Exam	Summative
9	Object detection concept: - bounding boxes and region proposals. - Practical application of bounding boxes for object detection.	Lecture (1)	Lecture On-campus	3,4,5	Discussion & Questions	Formative
		Lecture (2)	Lecture On-campus	3,4,5	Discussion & Questions	Formative
10	Introduction to image classification: SVM, RF and simple classifiers.	Lecture (1)	Lecture On-campus	1,3,5	Discussion & Questions	Formative
		Lecture (2)	Lecture On-campus	1,3,5	Discussion & Questions	Formative
11	Fundamentals of neural networks: Build and train a simple neural network	Lecture (1)	Lecture On-campus	4	Quiz#2	Summative
		Lecture (2)	Lecture On-campus	4	Discussion & Questions	Formative
12	CNN fundamentals: - Implement a basic CNN for image classification - Test different CNN architectures on sample images	Lecture (1)	Lecture On-campus	4,5	Discussion & Questions	Formative
		Lecture (2)	Lecture On-campus	4,5	Discussion & Questions	Formative

13	Advanced CNN Techniques: - Data augmentation, transfer learning, fine-tuning - Fine-tune a pre-trained CNN on custom data	Lecture (1)	Lecture On-campus	4,5	Discussion & Questions	Formative
		Lecture (2)	Lecture On-campus	4,5	Project, final exam	Formative
14	Object Detection with CNNs - CNN-based object detection models (YOLO, SSD) - Use a CNN-based model to detect objects in various images	Lecture (1)	Lecture On-campus	3,4,5	Discussion & Questions	Formative
		Lecture (2)	Lecture On-campus	3,4,5	Project, final exam	Summative
15	Comprehensive Review & Final Project Presentations	-	Lecture On-campus	2,3,4,5	Project	Summative
		-	Lecture On-campus	2,3,4,5	Project	Summative
16	Final Exam	Lecture (1)	Lecture On-campus	1-5	Final Exam	Summative

Course Evaluation Time and Weight

Assessment Tool	Weight %	Expected Due Date
Quizzes	10%	Week 5, 11
Mid Exam	25%	Week 8
Project	25%	Week 15
Final Exam	40%	Week 16

Course Policies

Item	Policy
Quizzes	- No makeup quizzes. 2 quizzes are given. - Each Quiz is out of 5.
Exams	The format of the exams is generally (but NOT always) as follows: General Definitions, Multiple-Choice, True/False, Analyze a Problem, Essay Questions, etc.
Makeup Exams	Makeup exam should not be given unless there is a valid excuse.
Drop Date	Last day to drop the course is according to the University regulations.
Academic Honesty	- Cheating or copying exams or quizzes is an illegal and unethical activity. - Standard MEU policy will be applied. - All graded assignments must be your own work (your own words).
Attendance	- Excellent attendance is expected. - MEU policy requires the faculty member to assign ZERO grade if a student misses 10% of the classes. - Sign-in sheets will be circulated when necessary. - If you miss class, it is your responsibility to find out about any announcements or assignments you may have missed.
E-Learning	It is your responsibility to check the course's eLearning website regularly (at least once every 2-3 days) for announcements and material.
Workload	The students should expect to spend 6 hours per week as an average workload. (more details are given below in JNQF Hours calculations)
Graded Exams	Instructors should return exam papers graded to students within one week after the exam date.
Participation	- Participation in and contribution to class discussions will affect your final grade positively. Raise your hand if you have any questions. - Making any kind of disruption and (side talks) in the class will affect you negatively.
Assignment/ Projects	- Homework Assignments are as mentioned in the previous table. - Students organize themselves in teams of (2-3) students each. Each team will give at least one presentation regarding a selected case, delivering a fully working application. - More details about phases and topics of the projects will be given later.
Others	

Rubrics and Feedbacks (If Applicable)

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* JNQF Hours Calculation

Activity	Duration/ week	Times/ Semester	No. of Hours
Learning Activities			

Lectures and Seminars	2	15	30
Tutorials	0	0	0
laboratory	2	15	30
Activities for Assessment and Evaluation			
First Exam/ Midterm	1	1	1
Second Exam (If applicable)			
Final Exam	2	1	2
Quizzes	0	0	0
Self-Learning Activities			
Assignments/ Homework	4	4	16
Case Studies	4	2	8
Presentations	1	1	3
E-learning			
Extra Readings	6	12	72
Total Hours	162		
JNQF Hours	Total Hours/10 = 16.2		